ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Section A: objective questions (30) · Section B: alternating voltages and currents, RC and RL circuits, RLC circuits and resonance, silicon-controlled rectifiers, oscillators and the 555 timer, inductors, encoders, and AC machines (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
5. Use π = 3,142 where needed.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The RMS value of a sine wave with a peak value of 100 V is about …

A 63,7 V B 141,4 V C 70,7 V D 50 V

1.1.2 A signal with a frequency of 50 Hz has a period of …

A 0,02 s B 50 s C 0,2 s D 2 ms

1.1.3 The time constant of an RC circuit is …

A R ÷ C B R × C C C ÷ R D 1 ÷ RC

1.1.4 A capacitor is considered fully charged after about … time constants.

A 1 B 2 C 10 D 5

1.1.5 The opposition of an inductor to AC is called …

A inductive reactance. B capacitive reactance.
C resistance only. D conductance.

1.1.6 At resonance in a series RLC circuit, …

A X_L is zero. B the impedance is at its maximum.
C $X_L = X_C$ and the current is at its maximum. D the current is zero.

1.1.7 An SCR (thyristor) is switched on by a pulse on its …

A anode. B cathode. C gate. D collector.

1.1.8 A 555 timer wired to produce a continuous square wave is working as a/an …

A astable multivibrator. B monostable multivibrator.
C bistable latch. D comparator only.

1.1.9 Inductors connected in series …

A have a smaller total inductance. B behave like capacitors.
C block DC. D add their inductances.

1.1.10 The speed of a three-phase induction motor is mainly set by the supply frequency and the number of …

A brushes. B poles. C commutator segments. D fuses.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 In a purely resistive AC circuit the voltage and current are in phase.

1.2.2 In a purely inductive circuit the current leads the voltage by 90°.

1.2.3 Capacitive reactance decreases when the frequency increases.

1.2.4 Impedance is measured in ohms.

1.2.5 Once an SCR is conducting, removing the gate signal switches it off.

1.2.6 An LC oscillator produces a sine wave at its resonant frequency.

1.2.7 A rotary encoder can be used to measure the speed and position of a robot wheel.

1.2.8 A squirrel-cage induction motor needs brushes and a commutator.

1.2.9 The rotor of an induction motor turns slightly slower than the rotating magnetic field.

1.2.10 A variable frequency drive (VFD) is used to control the speed of AC motors.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Time for one complete cycle	A	peak value
1.3.2	Highest value reached by an AC waveform	B	stator
1.3.3	Total opposition to AC in a circuit with R, L and C	C	monostable
1.3.4	Frequency at which X_L equals X_C	D	slip
1.3.5	Unit of inductance	E	resonant frequency
1.3.6	Minimum current needed to keep an SCR conducting	F	absolute encoder
1.3.7	Timer mode that gives one pulse when triggered	G	holding current
1.3.8	Encoder that gives a unique code for every shaft position	H	henry
1.3.9	Difference between synchronous speed and rotor speed	I	impedance
1.3.10	Stationary part of an AC motor that carries the windings	J	period

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: AC THEORY, RC AND RL CIRCUITS

2.1 A sinusoidal voltage is given by v = 325 sin(314t) V. Calculate: (a) the peak value; (b) the RMS value; (c) the frequency; (d) the period. (6)

2.2 A 10 kΩ resistor is connected in series with a 100 µF capacitor to a 9 V DC supply through a switch.

2.2.1 Calculate the time constant. (2)

2.2.2 How long will it take for the capacitor to be fully charged? (2)

2.2.3 Sketch the charging curve of the capacitor voltage and label the time constant and 63% point. (3)

2.3 Calculate the inductive reactance of a 0,2 H inductor at 50 Hz. (3)

[16]

QUESTION 3: RLC CIRCUITS AND RESONANCE

3.1 A series circuit has R = 30 Ω, X_L = 80 Ω and X_C = 40 Ω, connected to a 230 V, 50 Hz supply.

3.1.1 Calculate the impedance. (3)

3.1.2 Calculate the current. (2)

3.1.3 State whether the circuit is inductive or capacitive, and why. (2)

3.2 Calculate the resonant frequency of a circuit with L = 10 mH and C = 1 µF. (4)

3.3 Describe what happens to the current and impedance of a series RLC circuit as the frequency is increased through resonance. (3)

[14]

QUESTION 4: SCRs, OSCILLATORS AND THE 555

4.1 Draw the symbol of an SCR, label its terminals and explain how it is switched on and off. (6)

4.2 Give TWO applications of SCRs. (2)

4.3 A 555 timer is connected as an astable multivibrator with R_A = 1 kΩ, R_B = 10 kΩ and C = 10 µF. Use $f = 1{,}44 \div ((R_A + 2R_B) \times C)$.

4.3.1 Calculate the output frequency. (3)

4.3.2 Name ONE robot application of this circuit. (1)

4.4 Differentiate between a sinusoidal and a non-sinusoidal oscillator, and give ONE example of each. (4)

[16]

QUESTION 5: INDUCTORS, ENCODERS AND AC MACHINES

5.1 Inductors of 4 mH and 12 mH are available. Calculate the total inductance when they are connected (a) in series and (b) in parallel (assume no mutual inductance). (4)

5.2 Explain how an incremental rotary encoder measures the speed of a robot's motor, and how two channels (A and B) show the direction of rotation. (5)

5.3 A four-pole, three-phase induction motor runs from a 50 Hz supply. Its rotor turns at 1 440 r/min.

5.3.1 Calculate the synchronous speed using $n_s = 120f \div p$. (2)

5.3.2 Calculate the percentage slip. (3)

5.4 Explain how a rotating magnetic field makes the rotor of an induction motor turn. (5)

5.5 Give TWO reasons why induction motors are widely used in industry. (2)

5.6 Explain how a variable frequency drive (VFD) changes the speed of an induction motor. (3)

[24]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER